

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA0621	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	Alli Daity Rakshitha	Reg. No.:	192311181
Section / Batch:	CSE 4 TH YEAR	Date:	29/07/2026

Instructions to Students

- Answer BOTH scenario-based questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions	Marks
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1 - Q2	100 (2 × 50)
GRAND TOTAL:		100 Marks	

SECTION A – SCENARIO BASED ASSESSMENT

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Q1 Brute Force 50 marks **Q2** Brute Force 50 marks 2/2 answered

A banking system uses Binary Search on sorted customer IDs, but frequent insertions disturb order.
a) Implement how Binary Search works.
b) Analyze the impact of unsorted data.

Your Code

```
1 n=int(input())
2 arr=list(map(int,input().split()))
3 key=int(input())
4 low=0
5 high=n-1
6 found=-1
7 while low<=high:
8     mid=(low+high)//2
9     if arr[mid]==key:
10         found=mid
11         break
12     elif arr[mid]<key:
13         high=mid+1
14     else:
15         high=mid-1
16 if found!=-1:
17     print("Element found at index",found)
18 else:
```

Input

```
5
10 20 30 40 50
30
```

Output

```
Element found at index 2
Binary search works only on
sorted data.
if the data is unsorted,the
result may be incorrect.
```

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Q1 Brute Force 50 marks **Q2** Brute Force 50 marks 2/2 answered

A university portal stores student IDs in an unsorted format due to continuous registrations. During exams, quick lookup of student IDs is required.
a) Implement how Linear Search works for retrieving student records.
b) Analyze best, average, and worst-case time complexities.
c) Justify whether sorting the dataset would improve performance.

Your Code

```
1 n=int(input())
2 arr=list(map(int,input().split()))
3 key=int(input())
4 found=-1
5 for i in range(n):
6     if arr[i]==key:
7         found=i
8         break
9 if found!=-1:
10     print("element found at index",found)
11 else:
12     print("element not found")
13 print("Best Case:o(1)")
14 print("Average Case:o(n)")
15 print("Worst Case:o(n)")
```

Input

```
5
12 45 23 67 89
23
```

Output

```
element found at index 2
Best Case:o(1)
Average Case:o(n)
Worst Case:o(n)
Sorting improves performance only
if multiple searches are
```

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Code of Conduct

I Alli Daity Rakshitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Question No.	Problem Understanding & Context Analysis (25%)	Application of Concepts (30%)	Analytical & Decision-Making Skills (20%)	Solution Design & Approach (15%)	Clarity, Justification & Presentation (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____